PAPER-II
PHYSICS
SECTION – I
Single Correct Choice Type

1. Slide callipers can measure the external and internal diameters as well as the depth of a hollow cylinder. This is sufficient to find its volume and hence density, since mass is known.

$\therefore$ **(a)**

2. $I = ne\pi r^2 v$ [here n = no. of protons per unit volume and v is velocity of proton]

$$\varepsilon_0 (2\pi r v).\ E = I$$

$$\therefore \quad E = \frac{I}{2\pi r \varepsilon_0 v}$$

$$\frac{1}{2} m v^2 = \text{eV}$$

$$v = \sqrt{\frac{2eV}{m}}$$

$$E = \frac{I}{2\pi r \varepsilon_0} \sqrt{\frac{m}{2\,\text{eV}}}$$

$\therefore$ **(a)**

3. $\sqrt{\frac{3RT}{M}} =$ velocity of sound

$$\sqrt{\frac{3 \times R \times 273}{M}} = 330$$

$$\sqrt{\frac{3R}{M}} = \frac{330}{\sqrt{273}} \approx 20$$

Variation of temperature with height $T = T_1 - \left(\frac{T_1 - T_2}{y_2}\right) y$

On differentiation $1 = -\frac{(T_1 - T_2)}{y_2} \cdot \frac{dy}{dT}$

$$V = \sqrt{\frac{3RT}{M}} = \frac{dy}{dt} \cdot \frac{dT}{dT} = \frac{dy}{dT} \cdot \frac{dT}{dt}$$

$$-\frac{20(T_1 - T_2)}{y_2} \int_0^t dt = \int_{T_1}^{T_2} \frac{dT}{\sqrt{T}}$$

t = 30 sec

$\therefore$ **(a)**

4. $dQ = dU + dW$

$dW = Pdv = 10^5\left(0.5\times10^{-3}\right) = 50\,\text{J}$

$dU = 500 - 50 = 450$

$450 = a\dfrac{dT}{T} = a\dfrac{\left(T - T_0\right)}{T}$

$\dfrac{450}{a} = 1 - \dfrac{T_0}{T}$

$\dfrac{T_0}{T} = 1 - \dfrac{450}{3000} = \dfrac{2550}{3000}$

$T = 1.17\ T_0$

$\therefore$ **(b)**

SECTION – II
Multiple Correct Choice Type

5. Total energy $= \dfrac{1}{2}\dfrac{q_0^2}{C}$

From LC circuits $L\dfrac{dI}{dt} + \dfrac{q}{c} = 0$

The solution of above equation is $q = q_0\cos\omega t\ \left(\omega = \dfrac{1}{\sqrt{LC}}\right)$

At an instant $t = \dfrac{T}{8}$

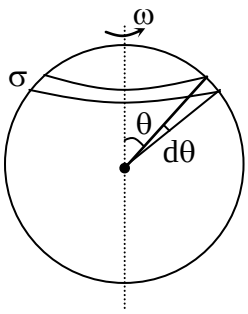
$q = q_0\cos\dfrac{2\pi}{T}\dfrac{T}{8} = q_0/\sqrt{2}$

$U_e = \dfrac{q^2}{2C} = \dfrac{{q_0}^2}{4C}$, $U_B = 50\%$ of total energy

Ratio is 1

$\therefore$ **(a)(b) and(d)**

6. Charge on differential strip $dq = \left(2\pi R\sin\theta\right)\left(Rd\theta\right).\sigma$



$dI = \dfrac{\omega dq}{2\pi} = \omega\sigma R^2\sin\theta d\theta$

Magnetic field $= \dfrac{\mu_0 dI\left(R^2\sin^2\theta\right)}{2R^3} = \dfrac{\mu_0\omega rR\sin^3\theta\, d\theta}{2}$

$B = \dfrac{\mu_0\omega\sigma R}{2}\int_0^{\pi}\sin^3\theta d\theta = \dfrac{2}{3}\mu_0\omega\sigma R$

Magnetic moment due to it

$\mu = \int dI\ \pi r^2 = 2\int_0^{\pi/2}\left(\omega\sigma R^2\sin\theta\right)\pi R^2\sin^2\theta d\theta = \dfrac{4}{3}\pi\sigma R^4\omega$

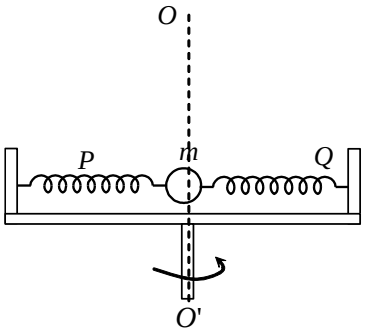
$\therefore$ **(a) and (b)**

7. $m\frac{d^2x}{dt^2} = -kx + m\omega^2 x$

$\frac{d^2x}{dt^2} = \left(-\frac{k}{m} + \omega^2\right)x$

Here $\omega_0 = \sqrt{\frac{k}{m} - \omega^2}$

$T = \frac{2\pi}{\sqrt{\frac{k}{m} - \omega^2}} = 0.70\text{sec}$



The oscillation will stop if $T = \infty$

It is possible when $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{28.8}{0.2}} = 12$ rad/sec

∴ **(a) and (c)**

8. $\tau = T_1 = (\mu R_1 + \mu R_2)r$

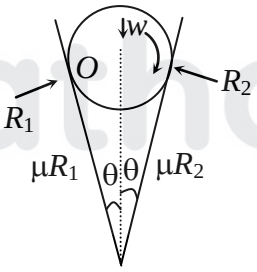
$w = (R_1 + R_2)\sin\theta + \mu(R_2 - R_1)\cos\theta$

In horizontal direction

$(R_2 - R_1)\cos\theta = \mu(R_1 + R_2)\sin\theta$

$w = (R_1 + R_2)(1 + \mu^2)\sin\theta$

$T_1 = \frac{\mu wr}{(1+\mu^2)\sin\theta}$



When sphere leaves contact with O

$R_1 = 0$, $T = \mu R_2 r$

$w = R_2(\sin\theta + \mu\cos\theta)$

$R_2 = \frac{w}{(\sin\theta + \mu\cos\theta)}$

In horizontal direction

$R_2\cos\theta = \mu R_2\sin\theta$

$\mu = \frac{\cos\theta}{\sin\theta}; T = \frac{(\cot\theta).wr}{\sin\theta + \frac{\cos^2\theta}{\sin\theta}} = wr\cos\theta$

∴ **(a), (b) and(c)**

SECTION – III
Reasoning Type

9. Momentum of the system of particles depends only on $\vec{F}_{ext}$ though momentum and kinetic energy of individual particle may change due to internal forces.

$\therefore$ **(d)**

10. $|\vec{R}_1| = \sqrt{A^2 + B^2 + 2AB\cos\phi}$

Now, $|\vec{R}_1| = |\vec{A}| = |\vec{B}| = 1$

So $\phi = 120^0$

$|\vec{R}_2| = \sqrt{A^2 + B^2 - 2AB\cos\phi}$

Now, $|\vec{A}| = |\vec{B}| = 1$

So $|\vec{R}_2| = \sqrt{3}$

$\therefore$ **(b)**

11. If images are formed by TIR, transmission do not take place. So total light is reflected back to form brighter image.

$\therefore$ **(a)**

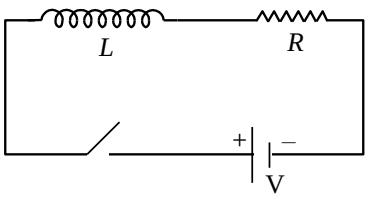
12. Kinetic energy = $h\nu$ - ϕ so K.E. $\propto \nu$

$$\text{K.E'} = \frac{h\nu}{2} - \phi \Rightarrow \text{K.E} \neq 2\,\text{K.E'}$$

$\therefore$ **(d)**

SECTION–IV
Comprehension Type
Passage-I

The rate of increase of energy is the difference between output power and power dissipated in resistor



$$\frac{d(E_{meg})}{dT} = VI - RI^2 = -R\left[I - \frac{V}{2R}\right]^2 + \frac{V^2}{4R} \leq \frac{V^2}{4R}$$

It is clear that rate of increase in maximum when $I = V/2R = 10/2\times 5$ = 1 amp

Only at $t = \infty$

$$I = \frac{V}{R} = \frac{10}{5} = 2 \text{ amp}$$

The power dissipation in resistor is

$$P = I^2 R = \frac{V^2}{R} = \left(1 - e^{-R/L.t}\right)^2$$

Fastest rate of change of energy becomes for both the inductor and resistor, when I^2 is changing most rapidly and this vacuums at $I = V/2R$

Put $I = V/2R$, we get $e^{-(R/L)t} = \frac{1}{2}$

$$t = 0.69\left(\frac{L}{R}\right) = 0.69 \times \left(\frac{500}{5}\right) = 69 \sec$$

13. **(a)**

14. **(b)**

15. **(a)**

Passage-II

$$dR = \frac{1}{\sigma(x)}\frac{dx}{A} = \frac{\sqrt{x}dx}{Al\sigma_0}$$

$$R = \int_0^l dR = \frac{1}{Al\sigma_0}\int_0^l \sqrt{x}dx = \frac{2\sqrt{l}}{3A\sigma_0}$$

$$I = \frac{v_0}{R} = \frac{3Av_0\sigma_0}{2\sqrt{l}}$$

$$J = \frac{I}{A} = \frac{3\sigma_0 v_0}{2\sqrt{l}}$$

$$E(x) = \frac{T}{\sigma(x)} = \frac{T}{\sigma_0 \frac{l}{\sqrt{x}}} = \frac{3v_0\sqrt{x}}{2(l)^{3/2}}$$

16. **(d)**

17. **(a)**

18. **(a)**

SECTION–V
Matrix–Match Type

1. **(A)** $\beta = \dfrac{\lambda D}{d}$

So on increasing D, β will increases

(B) $\beta = \dfrac{\lambda D}{d}$

So on increasing d, β will decrease

(C) $\lambda_w = \dfrac{\lambda air}{\mu_W}$ so $\beta_W = \dfrac{\lambda_{air} D}{\mu_W d}$ so, $\beta_W < \beta_{air}$

(D) Due to introduction of slab path difference at y = 0 is not zero

So all the fringes along with central fringe shift by same amount.

$\therefore$ **(A) – 1, 5; (B) – 2, 5; (C) – 2, 5; (D) – 3, 4**

2. **(A)** $(V_{AB})_v = 40\,\text{m/s} \;\therefore\; (S_{AB})_v = (40\times 2) = 80\,\text{m}$

$(V_{AB})_H = 0\,\text{m/s} \;\therefore\; (S_{AB})_H = 0\times 2 = 0\,\text{m}$

(B) $(V_{AB})_v = 40\,\text{m/s} \;\therefore\; (S_{AB})_v = (40\times 2) = 80\,\text{m}$

$(V_{AB})_H = 0\,\text{m/s} \;\therefore\; (S_{AB})_H = 0\times 2 = 0\,\text{m}$

(C) $(V_{AB})_v = 16\,\text{m/s}$, $(V_{AB})_H = 8\,\text{m/s}$

$\therefore\; V_{AB} = 8\sqrt{5}\,\text{m/s}$

$S_{AB} = 8\sqrt{5}\times 2 = 16\sqrt{5}\,\text{m}$

(D) $(V_{AB})_v = 24\,\text{m/s} \;\therefore\; (V_{AB})_H = 32\,\text{m/s}$

$\therefore\; V_{AB} = 40\,\text{m/s}$, $S_{AB} = 40\times 2 = 80\,\text{m}$

$(S_{AB})_H = 2\times 32 = 64\,\text{m}$

$\therefore$ **(A) – 2, 4, 5; (B) – 2, 4, 5; (C) – 1, 3; (D) – 2, 4**